Micrometrix®

SCM

Streaming Current Monitor and Dosing Controller · Prototype

Instruction Manual DRAFT

Document
SCM-UM-001

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Applies to firmware
0.9.57 and later

About this manual

This manual describes the installation, operation and routine maintenance of the Micrometrix SCM (Streaming Current Monitor). It is written for plant operators and maintenance staff. Engineering details of the internal wiring are kept in a separate document, *SCM Wiring Reference*, and are not needed for day-to-day use.

Keep this manual with the instrument. If the instrument changes hands, pass the manual on with it.

Conventions used in this manual

WARNING	Indicates a situation that could result in injury, chemical release, or damage to the process if the instruction is not followed.
CAUTION	Indicates a situation that could result in damage to the instrument or loss of data.
NOTICE	Information that helps you use the instrument correctly, with no safety implication.

On-screen buttons are shown as [ZERO]. Menu paths are written *Config → Dosing (PID) → Controller action*. Figures are numbered by chapter, for example **Figure 2-1**.

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1 Safety information

Read this section before installing or operating the instrument.

WARNING	**Chemical dosing.** In AUTO mode the instrument controls a chemical pump without operator action. An incorrect setpoint, an incorrectly set controller action, or a faulty sensor can cause overdosing or underdosing. Verify the controller action (section 6.2) in MANUAL mode before enabling AUTO on a live process, and confirm that the plant's own interlocks and alarms are in place.
WARNING	**Electrical.** Installation and connection of power, relay circuits and 4–20 mA loops must be carried out by a qualified person in accordance with local regulations. Isolate all supplies before opening the enclosure or touching terminals.
CAUTION	**Relay contacts** are dry contacts with a limited rating. Switch external loads through a suitably rated interposing relay or contactor. Fit a suppressor across inductive loads such as solenoid valves.
CAUTION	**Signal cable.** Do not extend, splice or re-route the sensor cable without consulting Micrometrix. The sensor signal is small and the cable is part of the measurement.
NOTICE	The instrument restarts itself automatically if its software ever stops responding, and returns the dosing output to the stopped (4 mA) state when it does. See section 11.

2 Product overview

2.1 Purpose

The SCM continuously measures **streaming current** — the net electrical charge carried by suspended particles in a water sample — and reports it in millivolts on a colour touchscreen. Streaming current is the most direct indicator of coagulant demand: a reading near zero means the particle charge has been neutralised and the dose is correct.

The instrument can also **control the dose**. A built-in PID controller drives a chemical metering pump through a 4–20 mA signal to hold the reading at a chosen setpoint. High and low alarm relays, a timed sensor clean cycle, a second 4–20 mA output that retransmits the reading, and data logging to a USB stick complete the instrument.

2.2 Features

Function	Description
Measurement	Streaming current, −1000 to +1000 mV, updated once per second, adjustable smoothing.
Relative / absolute reading	Absolute is the sensor value; relative is measured from the last zero point. Either can be the main display.
Dosing control	PI controller, direct or reverse acting, output limits, bumpless AUTO/MANUAL transfer, selectable power-up behaviour.
Alarms	High and low thresholds with relay outputs and on-screen indication; 10 mV release margin.
Clean cycle	Timed flush-valve relay with recovery hold; automatic on an interval or started manually.
Outputs	2 × 4–20 mA (pump command, reading retransmit); 3 relays (low alarm, high alarm, flush valve).
Trend	History of reading, dose and clean cycles, in seven windows from 1 minute to 24 hours.
Signal Health	Live sensor waveform, phase, motor speed and a health score.
Data logging	CSV rows to the instrument's own memory at a set interval (about 60 days kept), clean cycles in detail; export of the last hour, day, week or month to a USB stick.
Clock	Battery-backed real-time clock for timestamps.

Table 2-1 Instrument functions

NOTICE	**Two numbers, one reading.** Most plants control on the *relative* reading. The absolute reading is always shown in small text beneath the main number so it is never out of sight.

3 Front panel and screens

All operation is through the touchscreen. Five buttons down the left edge select the main screens; the bar across the top shows the model, the current screen and the time of day.

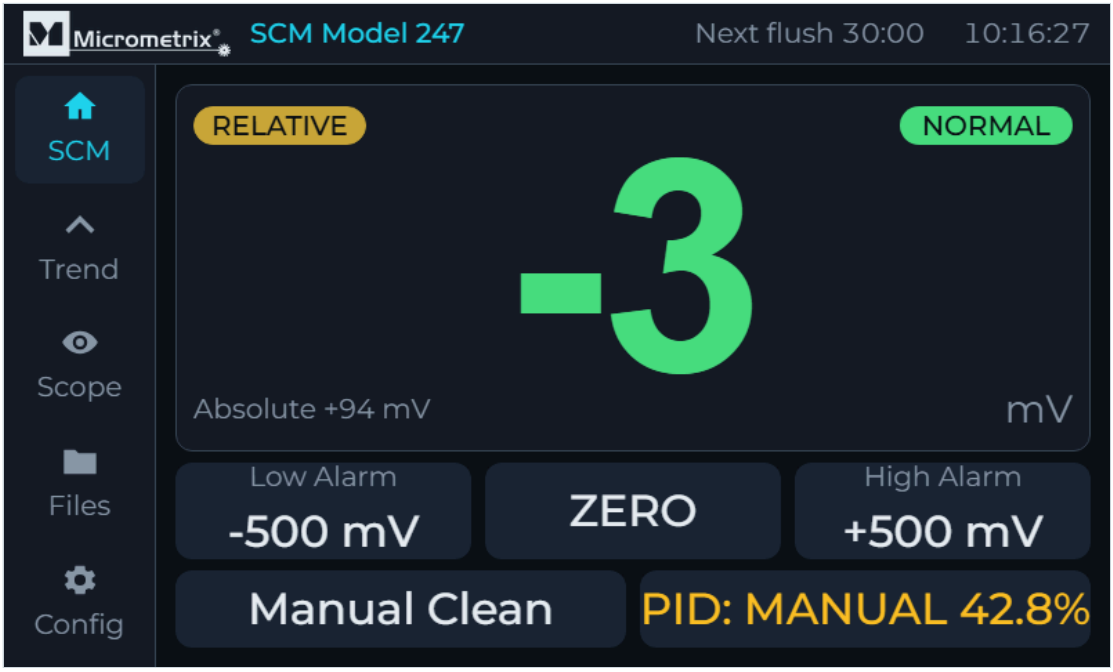


Figure 3-1 The SCM main screen

RELATIVE / ABSOLUTE — which reading the main number shows.

Status chip — NORMAL, LOW ALARM, HIGH ALARM, FLUSHING, RECOVERING, HEALTH LOW, NO FLOW, HOLD or SENSOR FAULT.

Low Alarm / High Alarm — tap to change a threshold.

PID — controller state and output; tap to switch AUTO / MANUAL.

Mode chip — RELATIVE (yellow) or ABSOLUTE (outline). Tap it to switch the main number.

Main number — streaming current in mV; green near zero, red in low alarm, blue in high alarm.

ZERO — make the current reading the new zero point.

Manual Clean — start a clean cycle now (shown when the clean feature is enabled).

NOTICE	In Figure 3-1 the PID button reads **NO DAC** because the dosing output module was not connected when the picture was taken. In normal service it reads **PID: MANUAL 0.0%** or **PID: AUTO 42.5%**.

3.1 Screen summary

Button	Screen	Purpose
SCM	Main reading	Day-to-day operation: reading, alarms, zero, dosing on/off, manual clean.
Trend	History	How the reading and the dose have moved over the last 15 minutes to 24 hours.
Scope	Signal Health	Condition of the sensor itself: waveform, phase, motor speed, health score.
Files	File Utilities	Logging status and export of logged data to the USB stick (section 8).
Config	Settings	Calibration, alarms, clean cycle, controller tuning, clock, system.

Table 3-1 Main screens

4 Daily operation

4.1 Reading the display

The main number is the streaming current in millivolts. Its colour gives the state at a glance: **green** within a few mV of zero, **grey** elsewhere in the normal band, **red** below the low alarm, **blue** above the high alarm. The status chip in the corner states the same thing in words.

The chip at the top left names the number on display: **RELATIVE** (measured from the last zero) or **ABSOLUTE** (the sensor value itself). Tap it to switch. The reading that is not on the large digits is shown in small text at the bottom left of the card, so both are always in view. The choice is kept through a power cycle.

4.2 Zeroing

Zeroing sets the reference point for the relative reading. Zero on a sample you regard as correctly treated.

1. Allow the reading to settle.
2. Tap ZERO and confirm.
3. The relative reading becomes 0. The absolute reading is unchanged.

CAUTION	Zeroing is not calibration. It moves the reference point only; it does not change how the sensor signal is converted to millivolts. Calibration (*Config → Sensor & alarms*) is normally performed once at installation.

4.3 Clean cycle

Streaming current sensors foul in service. The clean cycle opens the flush valve for a set period (*flush*), then holds the reading steady while the sensor recovers (*recover*). During both phases the status chip shows **FLUSHING** or **RECOVERING** with a countdown, alarms are held, and the dosing controller pauses so it does not react to the disturbance.

The held value is only what the *controller* sees. The Trend screen and the data log keep recording the live sensor reading throughout, so the sensor's recovery after a flush can be seen and judged (section 7.1).

- **Automatic:** set an interval in *Config → Cleaning*. The time to the next flush is shown in the top bar.
- **Manual:** tap Manual Clean on the main screen and confirm.

4.4 Entering values

Tapping any setting opens a numeric keypad. Enter the value, use ± for negative numbers, Clear to start again and Enter to save. Cancel leaves the setting unchanged.

Figure 4-1 — TO DO: screenshot of the keypad open on a setting

4.5 CAL DUE reminder

The instrument keeps track of when it was last calibrated. When the calibration interval set by your service provider has passed, a red **CAL DUE** badge flashes in the top bar on every screen.

- The instrument **keeps working normally**: measurement, alarms and dosing are not affected.
- Tap the badge to see when calibration fell due.
- Only your service provider can clear it, by recording the calibration on the Service & Calibration screen.

NOTICE	CAL DUE needs the clock. With no clock fitted or set, the instrument cannot tell how long it has been, and the reminder does not appear.

5 Alarms

Two thresholds are set from the main screen or from *Config → Sensor & alarms*. When the reading falls below the **Low Alarm** value or rises above the **High Alarm** value, the status chip and the colour of the main number change and the corresponding **alarm relay** closes, so an external annunciator or control system can register the condition.

An alarm releases automatically once the reading returns 10 mV inside the threshold. This margin prevents a reading resting exactly on the threshold from switching the relay repeatedly.

Figure 5-1 — TO DO: screenshot of the main screen in an alarm state

5.1 Sensor fault

The sensor signal is a 4–20 mA loop, and 4 mA is the bottom of the measuring range, not “no signal”. A broken cable or a failed transmitter gives 0 mA, which the instrument can tell apart from a genuine minimum reading. When the input falls below **3.6 mA** the instrument reports a **SENSOR FAULT** rather than a reading:

Item	While the fault is present
Main number	Four red dashes `----`, on the screen and on the LED display.
Status chip	**SENSOR FAULT**, in red. It takes priority over the alarm and clean-cycle states.
Low alarm relay	Closed, so the plant is notified.
Controller	Drops from AUTO to MANUAL at its minimum output and stays in MANUAL after the fault clears. AUTO cannot be selected while the fault is present.
Retransmit output	Driven to 3.5 mA, the conventional fault level, so a connected PLC or recorder can detect it.
Log	The *Fault* column reads 1.

Table 5-1 Sensor fault behaviour

The fault clears on its own once the input rises above 3.8 mA. Check the sensor cable and its connections first, then the transmitter and its power supply.

5.2 Signal Health alarm

Config → Sensor & alarms → Signal Health limit sets a Signal Health percentage (0 turns the alarm off). When the average Signal Health over a full minute falls below the limit, the status chip shows **HEALTH LOW** and the fault relay operates. The alarm releases when the one-minute average is 2% above the limit. A minute in which the sensor waveform could not be read neither trips nor clears it. Measurement and control continue; the alarm is a maintenance call, not a process alarm. A limit of 80–85% suits most installations.

5.3 Digital inputs

The instrument has two dry-contact inputs, terminals DI1 and DI2. Each has its own function, chosen in *Config → Inputs & outputs → Input 1* and *Input 2*, and its own *active when* setting, which decides whether a closed or an open contact counts as active (closed by default). The two are independent: for example a flow switch on Input 1 and a remote AUTO contact on Input 2.

Function	Behaviour
Off	The input is ignored.
Flow switch	An active contact means sample flow is present. Without flow the status chip shows **NO FLOW**, the reading and the controller output are held at their last values, the diagnostic relay operates, and log rows carry `NoFlow = 1`. Everything resumes when flow returns.
Remote hold	While the contact is active the reading and the controller are held and the chip shows **HOLD**. For a plant-side maintenance or sampling contact. Not a fault.
Remote clean	The contact becoming active starts a clean cycle, the same as Manual Clean.
Remote AUTO	The contact selects the controller mode: active = AUTO, inactive = MANUAL, transferred bumplessly. Acts when the contact changes, so the PID button on the screen still works in between.

Table 5-2 Digital input functions (either input)

If both inputs are given the same function they combine: **NO FLOW** is shown if either flow switch reports no flow, and **HOLD** if either hold contact is active. A remote clean or remote AUTO input acts on its own contact. An input is ignored for the first second after power-up, until its contact has been read twice.

5.4 Diagnostic relay

The fourth relay reports the conditions that call for a person rather than a process change: **sensor fault** (section 5.1), **Signal Health alarm** (5.2), **no flow** (5.3) and **calibration due** (4.5). *Config → Inputs & outputs → Diagnostic relay* selects its sense:

Setting	Relay
On fault	Energised while any of the four conditions is present; released otherwise. Same sense as the alarm relays.
Fail-safe	Energised while all is well; released on a fault, and also when the instrument loses power or stops running. Use this when the plant must be told about a dead instrument as well as a dead sensor.

Table 5-3 Diagnostic relay sense

6 Dosing control (PID)

The controller compares the reading with a **setpoint** (normally 0 mV) and adjusts the 4–20 mA signal to the metering pump to close the difference. The PID button on the main screen shows the mode and the present output as a percentage of pump range.

6.1 AUTO and MANUAL

Mode	Behaviour
MANUAL	Controller idle. The pump output is the value entered when AUTO was left, or set in *Config → Dosing (PID) → Manual dose output*. The instrument starts in this mode by default.
AUTO	Controller drives the pump to hold the setpoint. Transfer from MANUAL is bumpless: the controller starts from the output already set, so the pump does not step.

Table 6-1 Controller modes

Tap the PID button on the main screen; it opens the dose keypad with the present output filled in.

In	Keypad
MANUAL	Enter sets a new manual output. The AUTO key hands control to the controller (confirmed). Cancel changes nothing.
AUTO	The title reads *Switch to MANUAL*. Enter switches to MANUAL at the value entered; Cancel stays in AUTO.

Table 6-2 The PID button

6.2 Direct and reverse action

Reverse action (default): when the reading falls below the setpoint, dose *more*. This suits coagulant dosing of raw water, where the untreated reading is negative and coagulant raises it toward zero. **Direct** action is the opposite, for processes where more chemical drives the reading down. Set in *Config → Dosing (PID) → Controller action*.

WARNING With the wrong action the controller drives the reading *away* from the setpoint and opens the pump fully. Before enabling AUTO on a live process: in MANUAL, raise the pump output and observe which way the reading moves. Set the action accordingly.

6.3 Tuning parameters

Parameter	Default	Effect
PID setpoint	0.0 mV	The reading the controller aims for.
PID gain	1.000	Strength of response to error. Gain 1.000 moves the output 5% per 100 mV of error. Higher is faster but may oscillate.
PID integral time	60 s	How quickly a persistent error is worked off. Shorter is more aggressive; 0 disables integral action.
PID output min / max	0 / 100 %	Limits the controller will not exceed.
Mode at power-up	Manual 0%	Behaviour after a power interruption: start in MANUAL, resume last state, or start in AUTO.
Power-up manual output	0%	Pump output the instrument starts at when the mode above is Manual. 0% keeps the pump stopped until an operator acts (safest); a higher value resumes a base dose unattended. Shown only while the mode is Manual.

Table 6-3 Controller parameters

7 Trend and Signal Health

7.1 Trend



Figure 7-1 Trend screen, 30-minute window. Blue: reading (SC mV). Amber: controller output (PID %).

The tiles show the current, lowest and highest reading in the window. Tap **Window** to step through 1 min, 15 min, 30 min, 1 h, 4 h, 12 h and 24 h. Times along the bottom come from the instrument clock (with seconds in the 1-minute window).

The names across the top of the graph are also switches: tap **SC mV**, **PID** or **Flush / Recover** to hide that trace, and tap it again to bring it back. A hidden trace keeps its name, hollow and grey. The choice is kept through a power interruption, and a hidden trace is still recorded, so switching it back on shows its full history.

Trace	Meaning
SC mV (blue)	The reading, against the scale on the left.
PID % (amber)	Controller output, 0–100% of the chart height. The legend shows the present value, for example **PID 26.4 %**.
Clean cycle (square wave)	Runs along the floor while no cleaning is happening and steps up to a quarter of the chart height for the whole cycle: **red** while the flush valve is open, **green** while the sensor recovers.

Table 7-1 Trend traces

In AUTO the reading should sit near the setpoint and the *amber* line shows the process at work — a rising dose means the water is becoming harder to treat. Around a clean cycle the blue line shows the **live** reading, even though the main screen holds its value: the dip during the red part of the square wave and the climb back during the green part are the sensor recovering. If the reading has not settled before the square wave drops back to the floor, the recover time is too short.

7.2 Signal Health

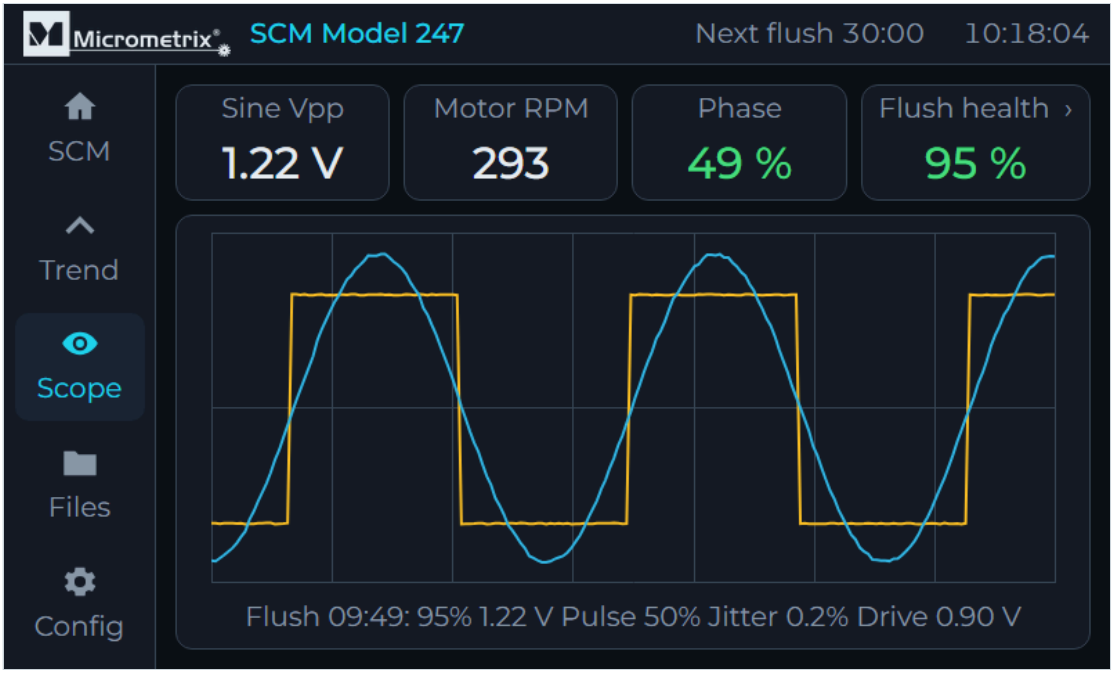


Figure 7-2 Signal Health. Sensor waveform (blue) against the drive reference (amber), 512 ms window.

This screen shows the sensor itself. Both traces are centred on the middle line, as on an AC-coupled oscilloscope.

Tile	Meaning
Sine Vpp	Peak-to-peak amplitude of the sensor signal, in volts, at the instrument input. Live.
Motor RPM	Sensor piston speed, from the drive reference. Known whenever the sensor runs, whatever the water. Live.
Phase	Where the signal peak sits within the drive pulse; 50% is centred and 45–55% is ideal. A worn drive shifts it. Needs a measurable signal, so it reads `---` on a near-neutral sample.
Flush health	0–100% grade of the sensor, measured **during the flush of the last clean cycle**: how well the signal peak is timed against the drive, how steadily the drive runs, and how clean and sine-like the signal is. Green above 85%, amber above 60%, red below. Tap the tile for its history (section 7.3).

Table 7-2 Signal Health tiles

The line under the plot says when the grade was taken and what it was, then the drive mechanics: **Pulse**, the drive's on-time as a percentage of its period (50% is symmetric); **Jitter**, how much the drive period changed since the previous sweep; and **Drive**, the amplitude of the reference wave.

Why the grade is taken during the flush. The signal's amplitude depends on the water: when the dose is right the sample is near its neutral point and a correctly working sensor produces almost no signal, so a grade taken then would say nothing about the sensor. The flush pushes raw, un-dosed water through the cell, and that is when the sensor shows what it can do. The instrument therefore grades the sensor during each clean cycle's flush, from 10 seconds in (once the water has changed over) to the end of the flush, and the result stands until the next flush. It is this flush grade that feeds the *Flush health* tile, the history (section 7.3) and the Signal Health alarm (section 5.2). Between flushes the live traces, Sine Vpp, motor speed and phase are still shown, because they are useful for a technician at the panel; a near-neutral sample makes the live phase read `---` and the line under the plot say so. Only when there is no drive wave at all does the plot show **No sensor signal** and the tiles read `--`: that is a cable, power or motor problem. An instrument with the clean cycle switched off takes no grade; set an interval (section 6.4) to get one.

A healthy sensor gives a clean, regular blue waveform locked to the amber reference. Ragged traces, hitches on the rising and falling edges, a drifting phase or a falling grade indicate a fouled or worn sensor (probe, piston or drive mechanics), usually

before the reading itself is obviously wrong. Clean the sensor first; if the grade does not recover, inspect the waveform and the mechanics.

7.3 Signal Health history

Figure 7-3 — TO DO: screenshot of the Signal Health history, 30-day view

Tap the **Signal Health** tile to see how the health of the sensor has changed over the last 7 days, 30 days or 60 days. A slow decline is the typical sign of a sensor that is fouling; a sudden step down points to an event at that time. Close the graph with ×. The same figures are in the *Health_pct*, *RPM* and *Sine_Vpp* columns of every log row (section 8.1), so an export carries them.

Item	Meaning
Blue line	Average health over each 12-hour period (00:00–12:00 and 12:00–24:00).
Grey line	Lowest 1-minute average in the same period. A grey line that falls away from the blue one means short drops rather than a general decline.
Right-most point	The period in progress, as it stands when the graph is opened.
Gaps	No record for that period: the instrument was switched off, or the sensor signal could not be read.
Line under the graph	The period in progress: hours so far, average and lowest health, share of the time the signal was readable, motor speed and its spread, sine amplitude.

Table 7-3 Signal Health history

One record is kept for each 12-hour period, for 60 days; after that the oldest is replaced. Records survive a power interruption. Each record is the average of the flush grades in its period (section 7.2), so a period with no clean cycle leaves a gap. The history needs the clock to be set. The scale runs from 80 to 100% and extends downward in steps of 20% when a value is lower.

8 Data logging and export

The **Files** button on the left opens File Utilities, a menu of three items: Data logging, Export configuration and Import configuration (section 8.2).

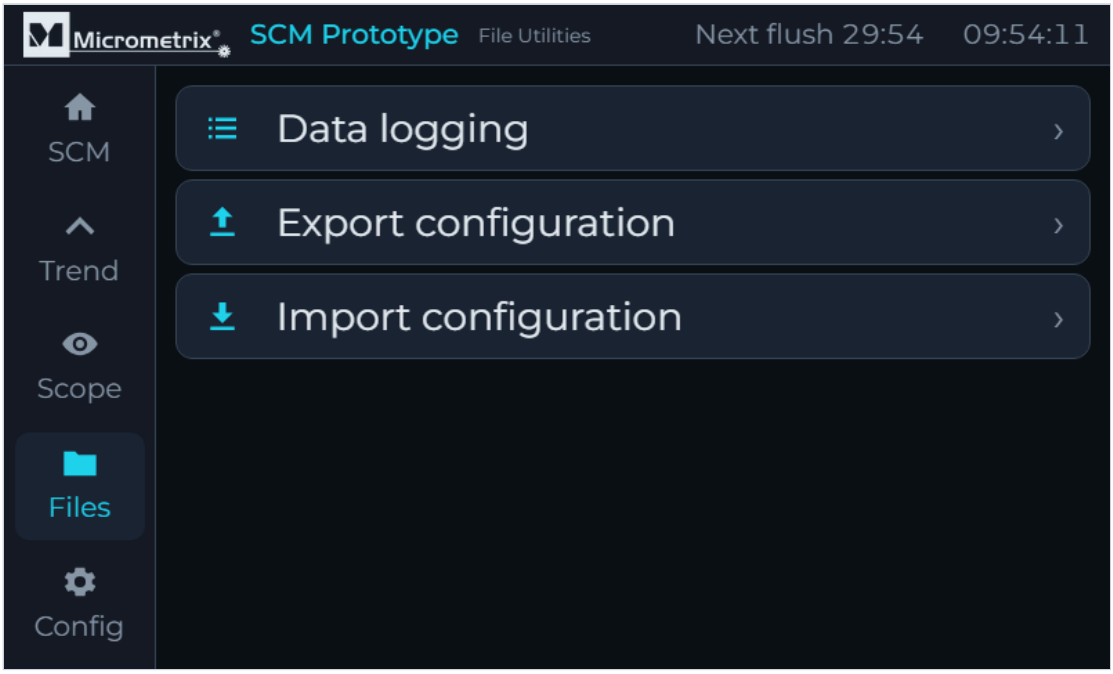


Figure 8-1 File Utilities (rendering from the virtual panel, firmware 0.9.37).

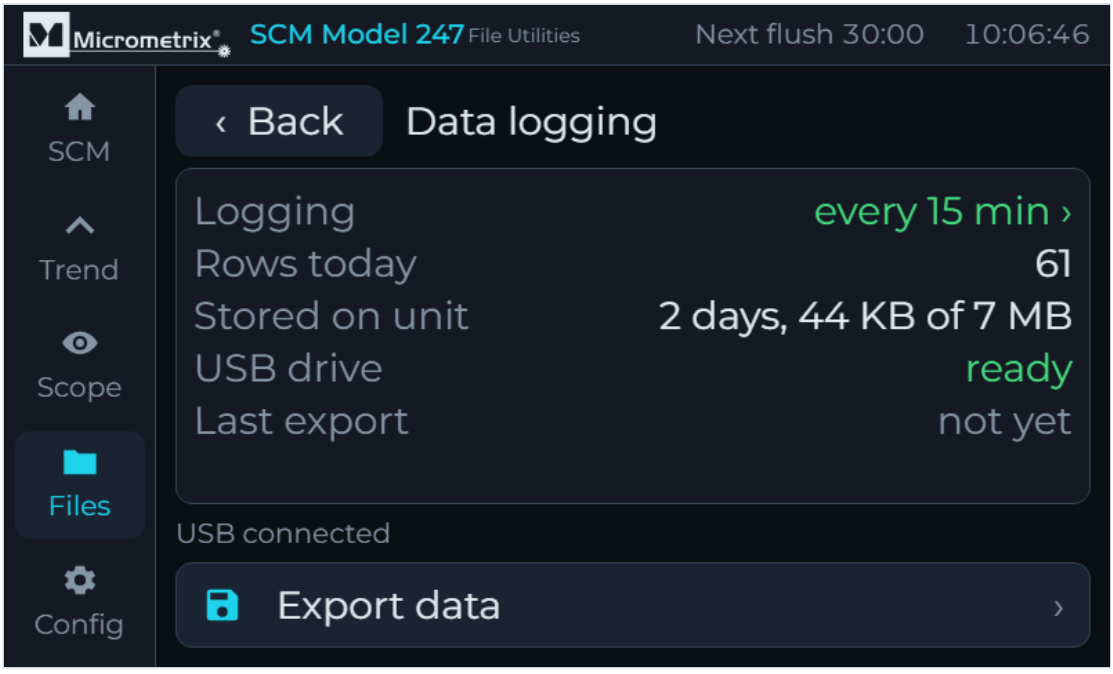


Figure 8-2 Data logging: status lines and the Export data row. Back returns to the menu.

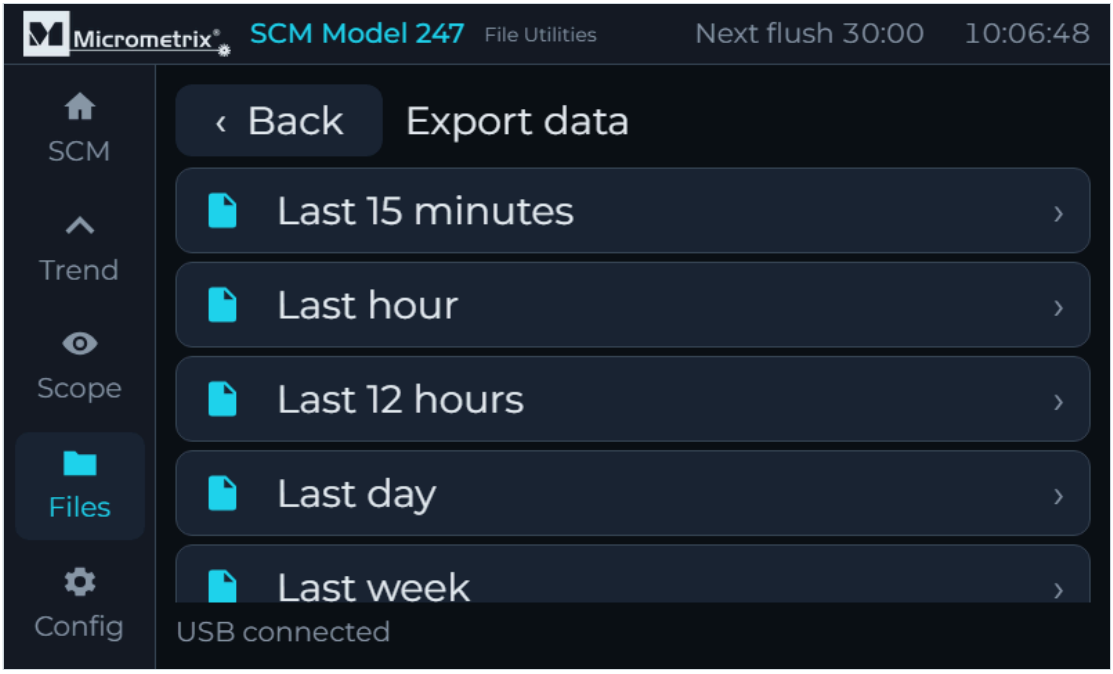


Figure 8-3 Export data: the windows, oldest-reaching last; the list scrolls. A progress bar and the line under it report progress and the result.

The instrument keeps its own log. One row is written to a file named for the day (for example `2026-09-17.csv`) every *Log interval* minutes (tap the *Logging* line on the Data logging page to change it), 15 by default, on the clock: at :00, :15, :30 and :45. A row is also written at power-up and at the start and end of each clean cycle. About 60 days are kept in the instrument's memory; the oldest day is removed as new days are added. Logging requires the clock to be running: a row without a trustworthy time is worthless. An instrument whose clock module is absent or has failed can still be set by hand (*Config → Display & clock*); it then logs until the next power-off, when the time is lost and logging pauses until the clock is set again. The *Time* row says *no clock chip* in that state.

During a clean cycle a row is written every 5 seconds to a second file for the day, `2026-09-17_clean.csv`, so each flush and recovery is captured in detail without crowding the main file.

A USB stick is not needed for logging, and a stick left in the port is not written to. The instrument writes to the stick only when asked: an export, or a configuration save (section 8.2). When that is finished the status line says *Safe to remove the stick*; at any other time the stick can simply be pulled. The **Stored on unit** line on the Data logging page shows how many days the instrument holds.

Which stick. Use a name-brand USB stick of 32 GB or less, formatted FAT32 (the size Windows formats as FAT32 by default). A stick that the instrument reports as *Drive seen, mount failed* and that a PC also struggles to read has a damaged file table; reformat it on the PC, and if it fails again, replace it. Sticks are the one part of the system that wears out.

Export data on the Data logging page opens a list of windows: Last 15 minutes, Last hour, Last 12 hours, Last day, Last week, Last 2 weeks, Last month and All stored data. Tapping one gathers that window into one file such as `Export_2026-09-17_LastHour.csv` on the stick, ready for a spreadsheet. The export runs in the background: the instrument keeps measuring and the screen stays usable while a progress bar fills and the line under it counts the rows, then reports the result. Everything stored can be some 25 000 rows and takes a minute or so on a typical stick; leave the stick in until the line says *Safe to remove the stick*. Clean-cycle detail rows (every 5 seconds during a flush and recovery) are included in time order; the *Clean* column reads `flush` or `recover` on them, so a spreadsheet filter on that column shows or hides them. The **Last export** line on the Data logging page shows the file name.

8.1 File format

Plain text, comma separated, one row per sample. Example with illustrative values:

```
Timestamp,Relative_mV,Absolute_mV,Output_pct,LowAlarm,HighAlarm,Clean,Health_pct,RPM,Sine_Vpp,Fault,NoFlow,HealthLow
2026-09-17 08:45:00,-6.0,91.0,40.3,0,0,idle,96.8,293.9,1.214,0,0,0
2026-09-17 09:00:00,-9.0,88.0,41.0,0,0,idle,96.6,293.8,1.212,0,0,0
2026-09-17 09:00:05,-148.0,-51.0,41.0,0,0,flush,96.1,293.8,1.215,0,0,0
2026-09-17 09:00:10,-212.0,-115.0,41.0,0,0,flush,96.0,293.7,1.216,0,0,0
2026-09-17 09:01:05,-97.0,0.0,41.0,0,0,recover,96.4,293.8,1.213,0,0,0
2026-09-17 09:01:55,-11.0,86.0,41.0,0,0,recover,96.7,293.9,1.212,0,0,0
2026-09-17 09:15:00,-8.0,89.0,40.6,0,0,idle,96.7,293.8,1.212,0,0,0
…
2026-09-17 23:59:59,,,,,,day average,96.7,293.8,1.213,,,
```

Column	Meaning
Timestamp	Date and time from the instrument clock.
Relative_mV	Reading relative to the last zero. Empty during a sensor fault.
Absolute_mV	Raw sensor reading. Empty during a sensor fault.
Output_pct	Controller output to the pump, 0–100%.
LowAlarm / HighAlarm	1 while that alarm is active, otherwise 0.
Clean	`idle`, `flush` or `recover`. In the main file a clean cycle appears as its first and last row; the 5-second rows are in the `_clean` file. During a clean cycle the readings are the live sensor values.
Health_pct	Average Signal Health since the previous row, 0–100%. Empty if the sensor signal could not be read.
RPM	Average sensor motor speed since the previous row.
Sine_Vpp	Average peak-to-peak amplitude of the sensor signal since the previous row, in volts at the instrument input.
Fault	1 while the sensor input is in fault (section 5.1), otherwise 0.
NoFlow	1 while the flow switch on DI1 reports no flow (section 5.3).
HealthLow	1 while the Signal Health alarm is active (section 5.2).

Table 8-1 Log file columns. Rows written by firmware before 0.9.21 end at *Clean*; before 0.9.24 at *Sine_Vpp*.

Daily Signal Health. Each day's file ends with one row marked `day average` in the *Clean* column, carrying that day's average Signal Health, motor speed and sine amplitude. It is the average of the day's `idle` rows, so clean cycles are left out, as in the Signal Health history. The row is added when the first row of the next day is written, so today's file receives it tomorrow. A day with no Signal Health data (no waveform input connected) receives no such row.

8.2 Configuration file

Export configuration writes the instrument's settings to `SCM_Config.txt` on the USB stick. It is a plain text file, one setting per line, that can be read and edited on a computer:

```
# MicroMetrix SCM Model 247 configuration
# GigaView firmware 0.9.45, exported 2026-10-07 09:08:39
alarm_low_mv = -500.000
alarm_high_mv = 500.000
pid_setpoint_mv = 0.000
input1_function = 1
log_interval_min = 15
```

Import configuration reads `SCM_Config.txt` from the stick and, after a confirmation, replaces the instrument's settings with those in the file. Use it to give a second instrument the same set-up, or to restore a known set-up. Each line is checked against

the range the keypad allows; a line that is not recognised or is out of range is skipped, and the result line under the menu reports how many settings were applied and how many lines were skipped.

NOTICE	The calibration of the sensor input and the two 4–20 mA outputs, the zero point and the service PIN belong to the individual instrument. They are written to the file for the record and are **not** read back by Import. An imported set-up therefore never disturbs an instrument's own calibration.

CAUTION	Import changes alarm limits, the controller setpoint and tuning, and the clean timings in one step. Put the controller in MANUAL before importing on a live process, and check the settings afterwards.

9 Settings reference

Tap Config on the menu rail. Settings are arranged in eight groups, listed with an icon. Tap a group to open its settings and Back to return to the menu; both lists scroll with a finger when they are longer than the screen. The firmware version is the last line of the menu.

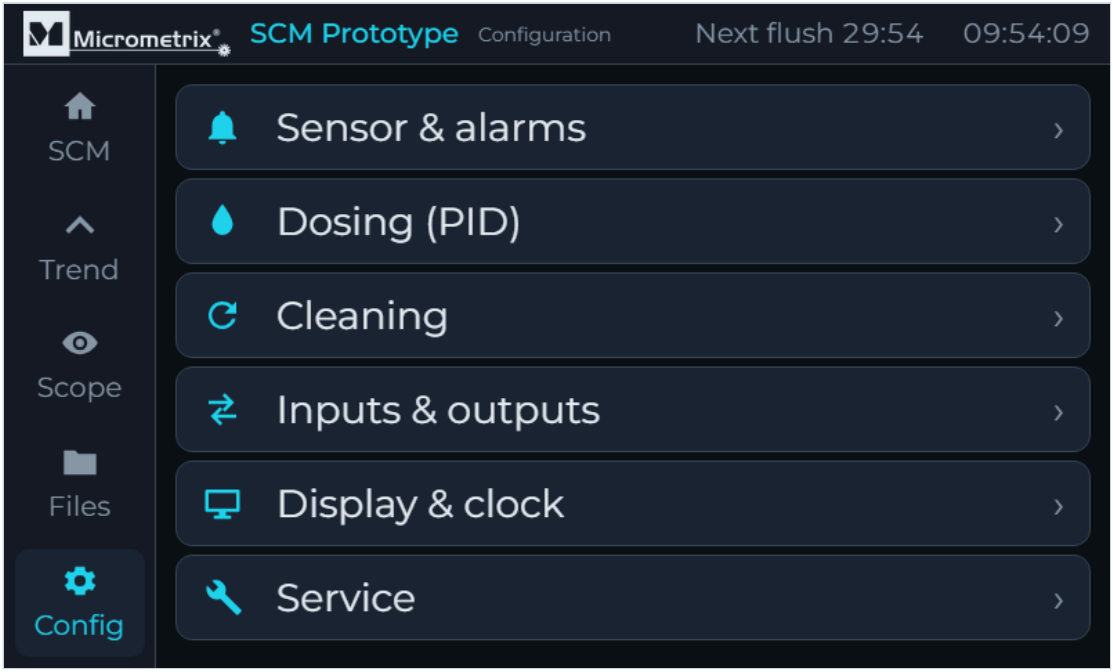


Figure 9-1 The Settings menu (rendering from the virtual panel, firmware 0.9.37).

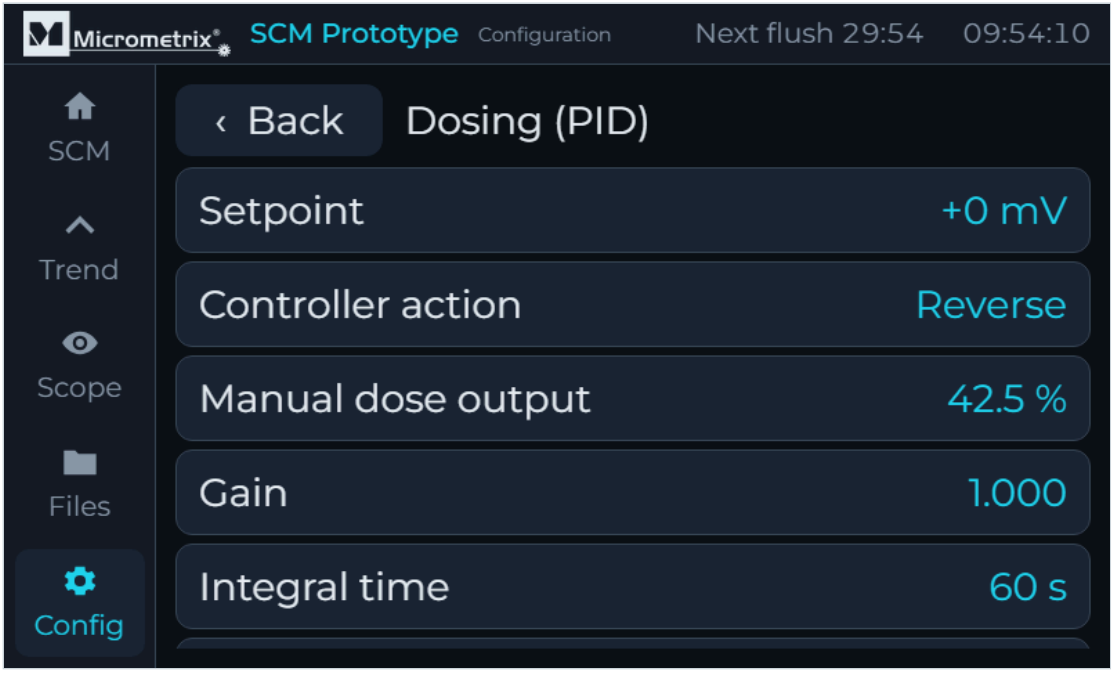


Figure 9-2 Inside a group. Back returns to the menu; the list scrolls.

NOTICE	The **Dosing (PID)** and **Cleaning** groups are listed only when those features are enabled (Service & Calibration screen, section 13.1).

Group	Setting	Function
Sensor & alarms	Low alarm / High alarm	Alarm thresholds (also on the main screen, section 5).
	Signal Health limit	Signal Health % below which the diagnostic relay operates (section 5.2); 0 = off.
	Averaging	Number of samples smoothed. Higher is steadier, slower.
	Input @ 4 mA / 20 mA	Two-point calibration of the sensor input. Installation setting.
Dosing (PID)	Setpoint, Controller action, Manual dose output	What an operator touches. Section 6.
	Gain, Integral time	Controller tuning (section 6.2).
	Output minimum / maximum	Pump output limits.
	Mode at power-up	Section 6.3.
Cleaning	Interval	Hours between automatic cycles; 0 = off.
	Flush time	Seconds the flush valve is open.
	Recover time	Seconds the reading is held after the valve closes.
Inputs & outputs	Digital input 1 / 2	Each: Off, Flow switch, Remote hold, Remote clean or Remote AUTO (section 5.3).
	Input 1 / Input 2 active when	Contact closed or Contact open counts as active.
	Diagnostic relay	On fault, or fail-safe (section 5.4).
	Alarm relays	**On alarm**: the low and high alarm relays are energised while their alarm is active. **Fail-safe**: energised while all is well and released on alarm, on power loss, or if the instrument stops. The flush valve relay is never fail-safe.
	Reading output module	Read-only: which reading-output board answered on the I2C bus, **GP8630N @ 0x5C** (DIP switch 1-0-0) or **GP8302 @ 0x58**. The GP8630N is tried first.
	Reading out / Dose out @ 4 mA / 20 mA	Calibration of the retransmit output and the pump output. Installation settings. The reading-out codes are 0–65535 for the GP8630N, 0–4095 for the GP8302; each board keeps its own pair.
Wiring	Read-only	One row per connection on the controller board: the pin, what is wired to it and why. The same facts as the wiring reference, readable at the panel.
Device manual	—	Shows a QR code for the web page with this manual and a browser simulation of the instrument. Scan it with a phone camera.
Display & clock	Main readout	Relative or Absolute as the main number. Tapping the RELATIVE / ABSOLUTE chip on the main screen does the same. Kept through a power cycle.
	Display range	±1000, ±100.0 or ±10.00 — moves the decimal point to match site convention. The measurement is unchanged.
	Brightness	10–100%. Never fully dark.
	Date / Time	Set the instrument clock.
Service	—	Opens the Service & Calibration screen (section 13.1). Changes there need the service PIN.

Table 9-1 Settings by group

The **log interval** is set under Files → Data logging: tap the *Logging* line (section 8). All settings in the table can be saved to and restored from a file (section 8.2).

10 Installation and connections

WARNING Connections must be made by a qualified person with all supplies isolated. See section 1.

Figure 10-1 — TO DO: end-user connection diagram of the enclosure and its terminals (sensor, pump loop, retransmit loop, alarm and valve contacts, USB, power). Pin-level detail remains in the *SCM Wiring Reference*.

Connection	Notes
Streaming current sensor	Supplied sensor and cable. Do not extend without advice.
Metering pump, 4–20 mA	Command to the pump. 4 mA = stopped, 20 mA = full rate.
Retransmit, 4–20 mA	The reading as a loop signal. 4 mA = −1000 mV, 20 mA = +1000 mV.
Low alarm / High alarm relays	Dry contacts; closed while the alarm is active.
Flush valve relay	Dry contact; closed during the flush phase.
Diagnostic relay	Dry contact; sensor fault, Signal Health alarm, no flow or calibration due (section 5.4).
Digital inputs DI1, DI2	Dry contact to common, each: flow switch, remote hold, remote clean or remote AUTO (section 5.3). Volt-free contacts only.
USB port	Export stick.
Power	*TO DO — specification of the finished enclosure supply.*

Table 10-1 External connections

11 Troubleshooting

Symptom	Likely cause and action
PID button reads **NO DAC**	Dosing output module not detected. Check its wiring and supply. The controller does not run until it is found; detection is retried every second.
NO FLOW, reading not moving	A flow switch on a digital input reports no sample flow. Reading and controller are held until flow returns (section 5.3). If flow is present, check the switch and that input's *active when* setting.
HEALTH LOW	Signal Health has been under its alarm limit for a minute (section 5.2). Run a clean cycle; if it does not recover, clean or inspect the sensor.
SENSOR FAULT, number shows `----`	Sensor input under 3.6 mA: broken or disconnected sensor cable, or the transmitter has no power. The controller has stopped dosing (section 5.1).
Reading fixed, grey; Signal Health `--`	The 4–20 mA input is present but the sensor waveform is not. Check the waveform cable and that the sensor motor runs.
Signal Health falling over days	Sensor fouling. Tap the Signal Health tile to see the history (section 7.3). Run a clean cycle; if health does not recover, clean or inspect the sensor.
Alarm relay switching repeatedly	The reading is noisy around a threshold. Increase Averaging.
Clock shows `--:--:--` or blinks red	Clock module not responding. Dashes: never seen since power-up. Blinking: was working, has stopped. Check connection and battery. Until it is fixed the clock can be set by hand after each power-up (section 8). Logging pauses until the clock returns.
Logging paused / Rows today stays 0	Requires a running clock and somewhere to write. **Stored on unit: not available** means the instrument's own memory could not be used (service item); with a USB stick fitted, logging continues on the stick.
Export writes nothing	"No log rows in the last ..." means no rows fall in that window yet; choose a wider one. A stick that mounts but cannot be written to reports a USB write error — try another stick.
Files on the stick show the wrong date	The instrument clock is not set. Set *Config → Display & clock → Date* and *Time*; files written afterwards carry the instrument's date and time.
Export file contains only the header	No log rows fall in the chosen window yet. Wait for the next interval or choose a wider window.
Controller drives the pump the wrong way	PID action incorrect for the process. Switch to MANUAL, change *Config → Dosing (PID) → Controller action* (section 6.2).
Signal Health plot jitters, "No sensor signal"	No usable sensor signal; the jitter is noise on the unconnected input. Check the sensor cable and that the sensor motor runs.
CAL DUE flashing in the top bar	The calibration interval has passed. The instrument keeps working normally. Contact your service provider to have it calibrated.
Instrument restarted by itself	The watchdog restarts the instrument if its software stops responding and returns the pump output to stopped. If it recurs, note the time and contact support.

Table 11-1 Fault finding

12 Specifications

Item	Value
Measurement range	−1000 to +1000 mV (displayable as ±100.0 or ±10.00)
Display update	1 per second
Display	7-inch colour touchscreen, 800 × 480
Analog outputs	2 × 4–20 mA (pump command, reading retransmit)
Relay outputs	4 (low alarm, high alarm, flush valve, diagnostic)
Digital inputs	2, volt-free contact; flow switch, remote hold, remote clean or remote AUTO
Sensor input	4–20 mA; below 3.6 mA reported as a sensor fault
Control	PI, direct or reverse acting, output limits, bumpless transfer
Trend memory	24 hours; seven windows from 1 minute
Data logging	One CSV file per day in the instrument's memory, about 60 days; 1–60 minute interval plus 5 s during clean cycles; export by hour, day, week or month to USB
Configuration file	Plain text on USB; export and import of operating settings
Clock	Battery-backed real-time clock
Settings retention	Non-volatile; retained through power loss and firmware updates
Power	*TO DO*
Enclosure / environment	*TO DO*

Table 12-1 Specifications

13 Service and support

For technical support, spare sensors and calibration services, contact Micrometrix.

TO DO — company address, telephone, e-mail, web address; warranty statement.

13.1 Service & Calibration screen (service providers)

Open it from *Config* → *Service*, or by tapping the CAL DUE badge. Anyone can read it; nothing on it changes without the **service PIN**.

Control	What it does
Unlock / Lock	Enter the service PIN to unlock. The screen locks again after 5 minutes without use, or as soon as you leave it. Five wrong PINs lock the keypad for a minute.
Mark calibrated	Records today as the last calibration, clears CAL DUE and starts the next interval.
Interval	Days between calibrations: 90 is 3 months, 180 is 6 months, 0 turns the reminder off.
Relay drive	Which way the relay board is driven: **Active low** (the usual opto-isolated modules) or **Active high**. A property of the hardware, set once at installation; all four relays change together. Not changed by a factory reset.
PID / Clean	Show or hide the dosing controller and the clean-cycle feature for this installation. Hiding PID also stops it.
Change PIN	Sets a new service PIN, 4 to 6 digits.
Factory reset	Restores every setting to factory values and marks calibration due. The service PIN and interval are kept.

Table 13-1 Service screen controls

With a USB stick fitted, each service action is recorded in `service_log.csv` on the stick. The PIN itself is never recorded.

Before you call

Please have ready: the firmware version (shown at the foot of the Config menu), a description of the symptom, and, if possible, a recent export file from the USB stick. A photograph of the screen is often the fastest way to describe what you are seeing.

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